

BEAMDATA.AI · PROTOTYPE PROJECT · SEPTEMBER 2026

Arabic Language Coach



An AI chatbot that coaches Arabic-speaking learners in English — correcting mistakes in real time and explaining every one in Arabic.

TEAM 4

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Current Challenges

Arabic speakers can't get real English conversation practice — **with corrections they actually understand.**



Learners

Arabic-speaking English students

- ✗ No partner to practice English conversation with
- ✗ Corrections come in English only — the "why" never lands
- ✗ Embarrassment freezes practice before it starts



Teachers & Tutors

Classroom & online instructors

- ✗ Re-correcting the same Arabic-L1 errors, student after student
- ✗ Class time can't give every learner real practice turns



Administrators

Programs & institutions

- ✗ 1:1 conversation coaching doesn't scale on a budget
- ✗ No consistent measure of spoken progress

Market Analysis

Where does our solution fit — and what's missing?

The market: digital language learning is large and established — AI conversation tutoring is its fastest-growing segment.

Duolingo

✗ Gamified drills — no live conversational coaching or real-time correction

ELSA Speak

✗ Pronunciation drills only — English-only feedback, no Arabic explanations

Cambly / italki

✗ Real human tutors — \$15–30/hour plus scheduling friction

Generic AI chatbots

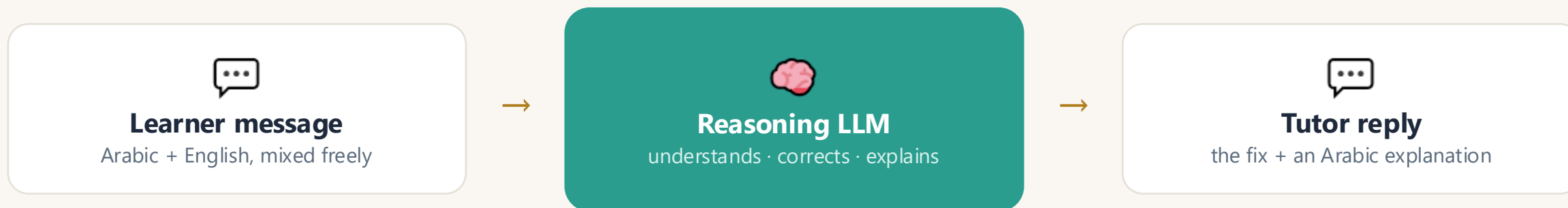
✗ Free-form chat — no lesson structure, no Arabic-L1 error focus

MARKET GAP

Nobody offers a structured bilingual coaching chat that truly code-switches Arabic ↔ English — targeting the mistakes Arabic speakers actually make, and explaining them in Arabic.

Proposed Solution

Arabic Language Coach is a bilingual AI chatbot that runs a complete, structured English lesson through conversation. The learner chats naturally — **mixing Arabic and English in the same message** — while a reasoning model carries every turn: it reads what was written, decides what to correct next, and replies with the fix plus a clear Arabic explanation. Mistakes get corrected the moment they're typed.



One model, one pipeline — no speech systems in between means lower latency and lower cost per lesson.

IN ONE LINE

“A patient English tutor in your pocket — it chats in your language, never sleeps, and costs cents per lesson.”

Bilingual Chat Conversation

01

No language toggle — follows Arabic and English mixed inside the same message, even mid-sentence

02

Switches with you — explanations in Arabic, target practice in English

03

Handles how learners actually write — naturally and code-switched, not textbook-perfect

REAL OUTPUT — BILINGUAL

 0.59 s

PROMPT — ARABIC

أريد كوباً من الماء من فضلك: ترجم إلى الإنجليزية ترجمة طبيعية

MODEL OUTPUT

"I would like a glass of water, please."

EXPLANATION — BOTH LANGUAGES

«أريد» = "I would like" · «كوباً من الماء» = "a glass of water"

Polite tone — **"please"** kept at the end, as in the original

PROMPT #10 · 161 TOKENS · FINISH ✓

WHY IT MATTERS You start in the language you're comfortable in — not the one you're still learning.

Instant Correction, Explained in Arabic

01

Tuned to real Arabic-L1 errors — missing "to be",
dropped articles, "there is" vs "it has"

02

Corrects at the right moment — never over-correcting
fluent speech or inventing errors

03

Every fix ships with an Arabic "why" — warm tutor tone,
not clinical

REAL OUTPUT — EXPLANATION

⚡ 0.77 s

PROMPT — ARABIC

على الرغم :ترجم إلى الإنجليزية مع الحفاظ على المعنى والنبرة
من أن الرحلة كانت طويلة، فإنني استمتعت بكل لحظة فيها

MODEL OUTPUT

"Although the trip was long, I enjoyed every moment of it."

WORD-BY-WORD BREAKDOWN

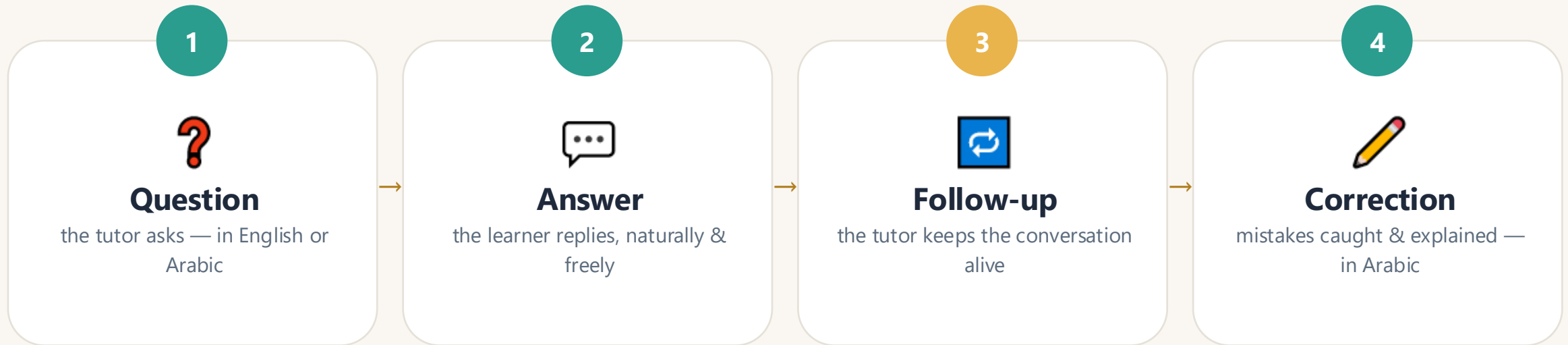
«على الرغم من أن» = "Although"
«الرحلة كانت طويلة» = "the trip was long"
«استمتعت بكل لحظة فيها» = "I enjoyed every moment of it"

PROMPT #12 · 192 TOKENS · MEANING & TONE PRESERVED

WHY IT MATTERS A correction you understand in your own language is a correction you remember.

The Coaching Loop

Every exchange follows one simple tutoring loop — repeated naturally throughout the conversation.



🔄 the loop repeats — turn after turn, lesson after lesson

SIMPLE BY DESIGN Four steps, one model, one conversation.

WHERE LEARNING HAPPENS Corrections arrive inside the conversation.

KEY FEATURE 4

Evidence-Based Model Choice



Three candidates evaluated on our own bilingual dataset — measured, not assumed.

METRIC	gpt-5.6-terra	Qwen3-8B (AWQ) 🏆 WINNER	ALLaM-7B
Success rate	100%	100%	100%
Avg latency	3.55 s	1.42 s	1.64 s
First token (TTFT)	1.51 s	0.02 s	0.05 s
Output speed	43 tok/s	111 tok/s	47 tok/s

🏆 **WINNER** Qwen3-8B-AWQ — the best results on our dataset · 2.5× faster responses · a fraction of the cost per lesson

* ALLaM-7B generated ≈ half the output length per request. 100 requests benchmarked per model.

WHY IT MATTERS In a chat lesson, first-token speed is the difference between a tutor that feels instant and one that feels stuck.

How Users Access It



PROTOTYPE FOCUS

Web Platform

<https://t10.aidc.nadir.sh/>

Sign in and chat. The full coaching lesson runs in the browser as a text conversation — nothing to install.

SCOPE & CONSTRAINTS Demo focuses on the **web platform** · needs internet only — **no microphone, no voice setup** · no voice mode in this prototype. **Voice is a natural future extension.**

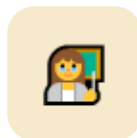
Benefits for All Stakeholders

The same three groups from the start — every pain point answered.



Learners

- ✗ No practice partner
- ↓ **Unlimited conversation practice — any time, zero embarrassment**
- ✗ English-only feedback
- ↓ **Corrections explained in Arabic — they actually stick**



Teachers & Tutors

- ✗ Same errors, re-corrected
- ↓ **Routine L1-error drilling handled by the AI**
- ✗ No time for 1:1 turns
- ↓ **Class time shifts to high-value coaching**



Administrators

- ✗ Coaching doesn't scale
- ↓ **1:1 coaching at cents per lesson**
- ✗ Progress unclear
- ↓ **Consistent, measurable progress tracking**

SEE IT LIVE

Demo

